In [ ]:
```
# EXPERIMENT 2 – NUMPY OPERATIONS
PIN NO.: 241UDS0103

 AIM
To study and practice NumPy operations including array creation, attributes,
indexing, slicing, broadcasting, arithmetic, statistical operations,
concatenation, reshaping, sorting, splitting, and retrieving selected data.
```

In [ ]:
```
## Step 1 – Install and Import NumPy
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2 – Create Arrays for Three Energy Sources
```

In [2]:
```
solar_output_241UDS0103 = np.array([410, 460, 435, 490])
wind_output_241UDS0103 = np.array([360, 395, 380, 420])
hydro_output_241UDS0103 = np.array([520, 510, 540, 555])

solar_output_241UDS0103, wind_output_241UDS0103, hydro_output_241UDS0103
```

Out[2]:
```
(array([410, 460, 435, 490]),
 array([360, 395, 380, 420]),
 array([520, 510, 540, 555]))
```

In [3]:
```
energy_data_241UDS0103 =
pd.read_csv("Experiment_2_241UDS0103_Renewable_Energy.csv")
energy_data_241UDS0103
```

Out[3]:

	Energy_Source	Week	Energy_Output_kWh
0	Solar	1	410
1	Solar	2	460
2	Solar	3	435
3	Solar	4	490
4	Wind	1	360
5	Wind	2	395
6	Wind	3	380
7	Wind	4	420
8	Hydro	1	520
9	Hydro	2	510
10	Hydro	3	540
11	Hydro	4	555

In [ ]:
```
## Step 3 – Check Array Properties
```

In [4]:
```
solar_output_241UDS0103.ndim
```

Out[4]:
```
1
```

In [5]:
```
wind_output_241UDS0103.shape
```

Out[5]:
```
(4,)
```

In [6]:
```
hydro_output_241UDS0103.size
```

Out[6]:
```
4
```

In [7]:
```
solar_output_241UDS0103.dtype
```

Out[7]:
```
dtype('int64')
```

In [8]:
```
solar_output_241UDS0103.itemsize, wind_output_241UDS0103.itemsize,
hydro_output_241UDS0103.itemsize
```

Out[8]:
```
(8, 8, 8)
```

In [ ]:
```
## Step 4 — Retrieve Data Using Indexing and Slicing
```

In [9]:
```
hydro_output_241UDS0103[0]
```

Out[9]:
```
np.int64(520)
```

In [10]:
```
np.unique(wind_output_241UDS0103)
```

Out[10]:
```
array([360, 380, 395, 420])
```

In [11]:
```
solar_output_241UDS0103[0:2]
```

Out[11]:
```
array([410, 460])
```

In [12]:
```
wind_output_241UDS0103[2:]
```

Out[12]:
```
array([380, 420])
```

In [13]:
```
hydro_output_241UDS0103[-3:]
```

Out[13]:
```
array([510, 540, 555])
```

In [ ]:
```
## Step 5 — Assign New Values Using Broadcasting
```

In [14]:
```
solar_output_new_241UDS0103 = solar_output_241UDS0103.copy()
solar_output_new_241UDS0103
```

Out[14]:
```
array([410, 460, 435, 490])
```

In [15]:
```
solar_output_new_241UDS0103 = np.append(
    solar_output_new_241UDS0103,
    [505, 475]
)
solar_output_new_241UDS0103
```

Out[15]:
```
array([410, 460, 435, 490, 505, 475])
```

In [16]:
```
solar_output_new_241UDS0103[4:6] = [515, 485]
solar_output_new_241UDS0103
```

Out[16]:
```
array([410, 460, 435, 490, 515, 485])
```

In [ ]:
```
## Step 6 — Arithmetic Operations
```

In [17]:
```
wind_output_241UDS0103
```

Out[17]:
```
array([360, 395, 380, 420])
```

In [18]:
```
wind_output_241UDS0103 + 25
```

Out[18]:
```
array([385, 420, 405, 445])
```

In [19]:
```
wind_output_241UDS0103[3] - 15
```

Out[19]:
```
np.int64(405)
```

In [20]:
```
wind_output_241UDS0103[0] * 2
```

Out[20]:
```
np.int64(720)
```

In [21]:
```
wind_output_241UDS0103[1:3] / 2
```

Out[21]:
```
array([197.5, 190. ])
```

In [ ]:
```
## Step 7 — Statistical Operations
```

In [22]:
```
solar_output_241UDS0103, wind_output_241UDS0103, hydro_output_241UDS0103
```

Out[22]:
```
(array([410, 460, 435, 490]),
 array([360, 395, 380, 420]),
 array([520, 510, 540, 555]))
```

In [23]:
```
np.sum(solar_output_241UDS0103)
```

Out[23]:
```
np.int64(1795)
```

In [24]:
```
np.max(solar_output_241UDS0103), np.min(solar_output_241UDS0103)
```

Out[24]:
```
(np.int64(490), np.int64(410))
```

In [25]:
```
print(
    np.mean(solar_output_241UDS0103),
    np.mean(wind_output_241UDS0103),
    np.mean(hydro_output_241UDS0103)
)
```

```
448.75 388.75 531.25
```

In [ ]:
```
## Step 8 — Combine All Energy-Source Data
```

In [26]:
```
solar_output_241UDS0103.ndim, wind_output_241UDS0103.ndim,
hydro_output_241UDS0103.ndim
```

Out[26]:
```
(1, 1, 1)
```

In [27]:
```
all_energy_output_241UDS0103 = np.concatenate([
    solar_output_241UDS0103,
    wind_output_241UDS0103,
    hydro_output_241UDS0103
])
all_energy_output_241UDS0103
```

Out[27]:
```
array([410, 460, 435, 490, 360, 395, 380, 420, 520, 510, 540, 555])
```

In [28]:
```
np.sum(all_energy_output_241UDS0103)
```

Out[28]:
```
np.int64(5475)
```

In [ ]:
```
## Step 9 — Reshape into a 3 × 4 Matrix
```

In [30]:
```
all_energy_output_241UDS0103.ndim, all_energy_output_241UDS0103.shape
```

Out[30]:
```
(2, (3, 4))
```

In [31]:
```
all_energy_output_241UDS0103 = all_energy_output_241UDS0103.reshape(3, 4)
all_energy_output_241UDS0103
```

Out[31]:
```
array([[410, 460, 435, 490],
       [360, 395, 380, 420],
       [520, 510, 540, 555]])
```

In [32]:
```
all_energy_output_241UDS0103.ndim, all_energy_output_241UDS0103.shape
```

Out[32]:
```
(2, (3, 4))
```

In [ ]:
```
## Step 10 — Sort the Array
```

In [33]:
```
all_energy_output_241UDS0103
```

Out[33]:
```
array([[410, 460, 435, 490],
       [360, 395, 380, 420],
       [520, 510, 540, 555]])
```

In [34]:
```
np.sort(all_energy_output_241UDS0103, axis=1)
```

Out[34]:
```
array([[410, 435, 460, 490],
       [360, 380, 395, 420],
       [510, 520, 540, 555]])
```

In [35]:
```
np.sort(all_energy_output_241UDS0103, axis=0)
```

Out[35]:
```
array([[360, 395, 380, 420],
       [410, 460, 435, 490],
       [520, 510, 540, 555]])
```

In [ ]:
```
## Step 11 — Split the Array
```

In [36]:
```
split_solar_241UDS0103, split_wind_241UDS0103, split_hydro_241UDS0103 =
np.split(
    all_energy_output_241UDS0103,
    [1, 2],
    axis=0
)
```

In [37]:
```
split_solar_241UDS0103
```

Out[37]:
```
array([[410, 460, 435, 490]])
```

In [38]:
```
split_wind_241UDS0103
```

Out[38]:
```
array([[360, 395, 380, 420]])
```

In [39]:
```
split_hydro_241UDS0103
```

Out[39]:
```
array([[520, 510, 540, 555]])
```

In [ ]:
```
## Step 12 – Retrieve Week 2 Output of All Energy Sources
```

In [40]:
```
week_2_output_241UDS0103 = all_energy_output_241UDS0103[:3, 1:2]
print(week_2_output_241UDS0103)
```

```
[[460]
 [395]
 [510]]
```

In [ ]:
```
## RESULT
Using NumPy, arrays were created and their properties checked. Indexing,
slicing, appending and updating values,
arithmetic and statistical operations, concatenation, reshaping, sorting,
splitting, and selecting Week 2 values were performed
on the renewable-energy dataset.
```